

PRAKTIKUM SISTEM OPERASI

**MODUL 3
(EXPLORASI XINU)**



Oleh:

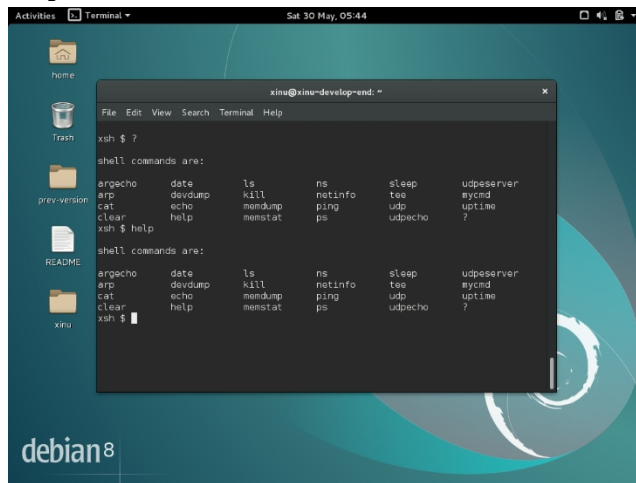
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(2311104057)

**PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK
DIREKTORAT KAMPUS PURWOKERTO
UNIVERSITAS TELKOM
2026**

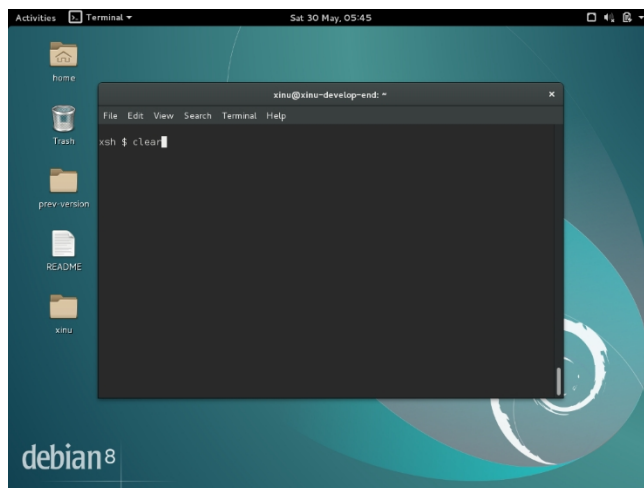
JURNAL 3

1. help dan ?



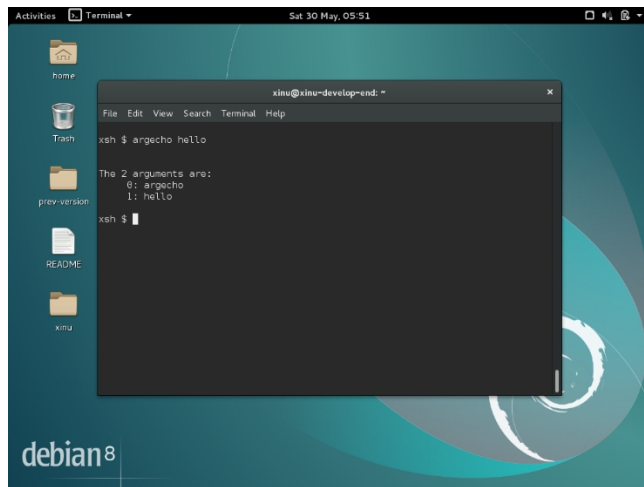
Menampilkan command yang tersedia di Xinu

2. clear



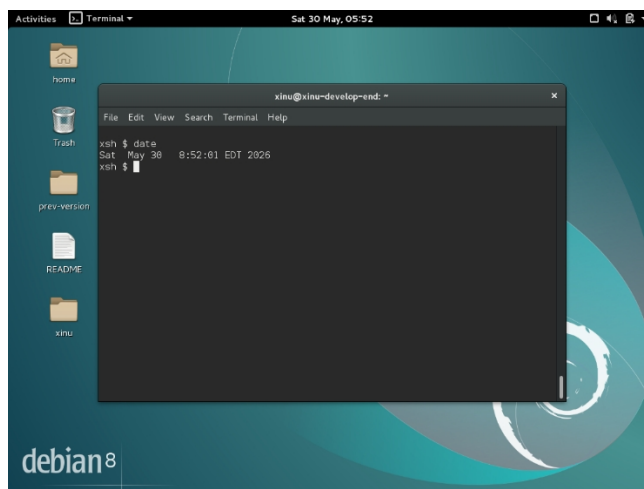
Membersihkan tampilan teks pada layar terminal.

3. argecho



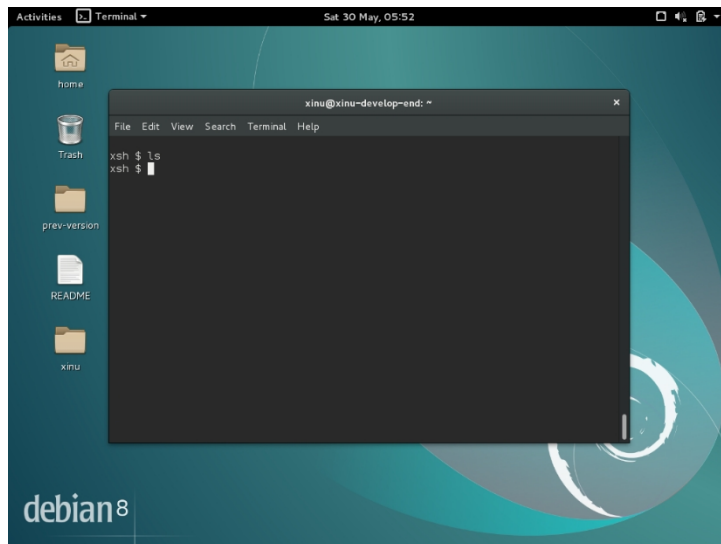
Mencetak dan menampilkan argumen baris perintah

4. date



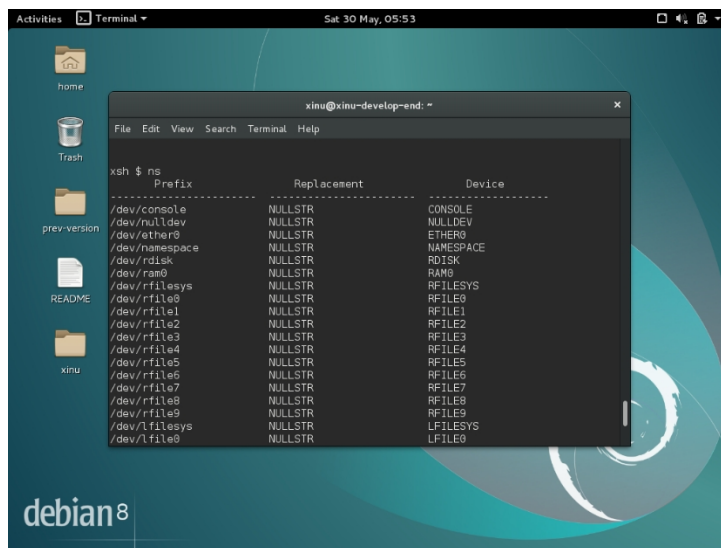
Menampilkan tanggal dan waktu system

5. ls



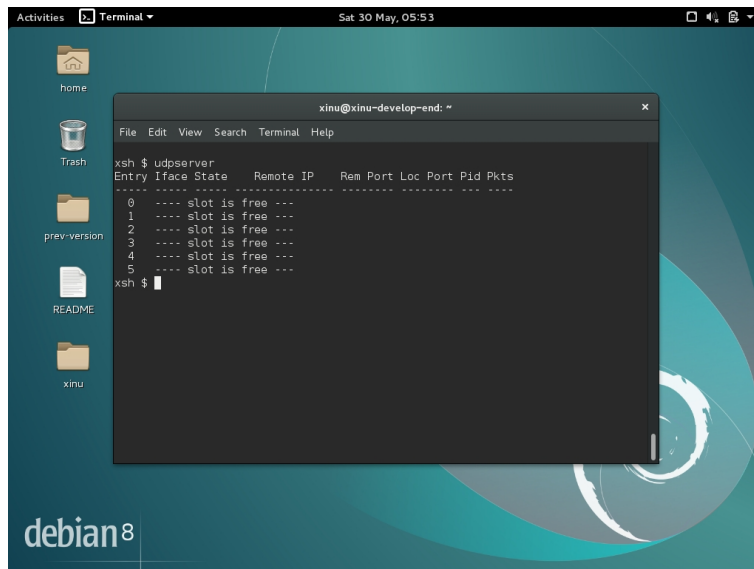
Menampilkan isi direktori

6. ns



menjalankan network server (server jaringan) bawaan

7. udpsvr

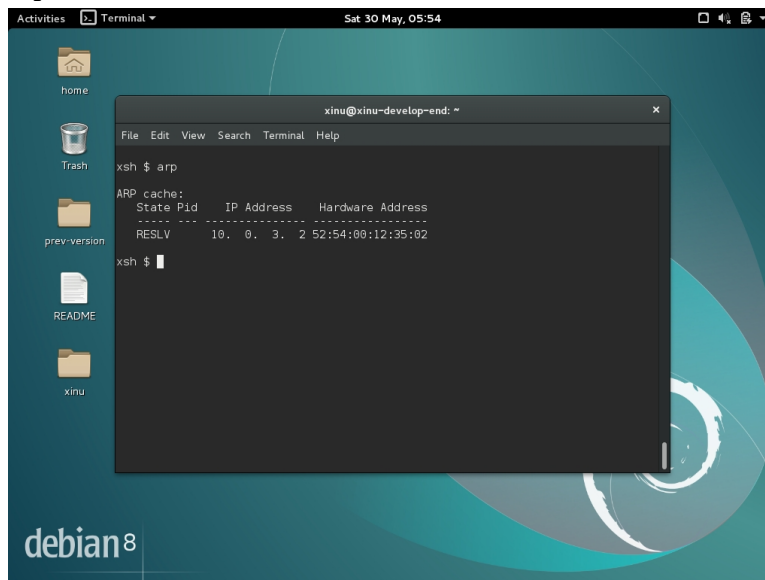


A terminal window titled 'xinu@xinu-develop-end: ~' is open on a Debian 8 desktop. The user has entered the command 'xsh \$ udpsvr'. The output shows a table with columns: Entry, Iface, State, Remote IP, Rem Port, Loc Port, Pid, and Pkts. The table lists six entries, all with the state 'slot is free'.

```
xinu@xinu-develop-end: ~  
File Edit View Search Terminal Help  
xsh $ udpsvr  
Entry Iface State Remote IP Rem Port Loc Port Pid Pkts  
-----  
0 ---- slot is free ---  
1 ---- slot is free ---  
2 ---- slot is free ---  
3 ---- slot is free ---  
4 ---- slot is free ---  
5 ---- slot is free ---  
xsh $
```

menjalankan server UDP sederhana

8. arp

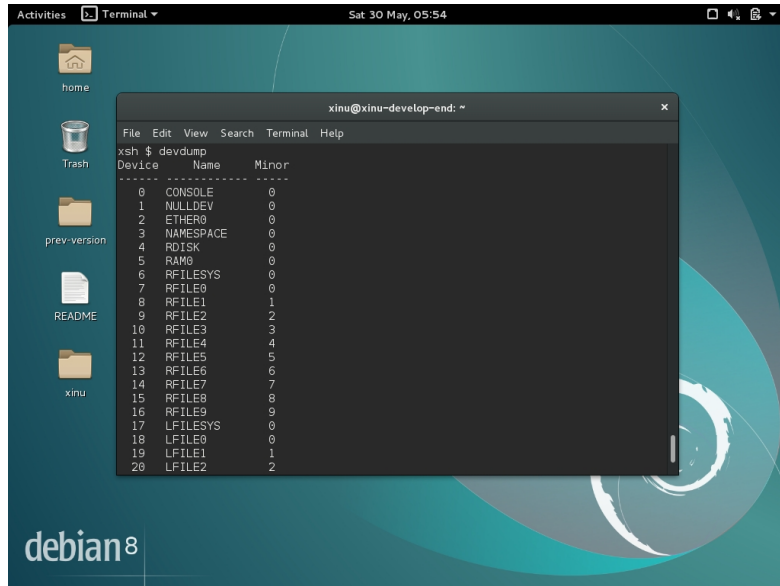


A terminal window titled 'xinu@xinu-develop-end: ~' is open on a Debian 8 desktop. The user has entered the command 'xsh \$ arp'. The output shows the ARP cache table with columns: State, Pid, IP Address, and Hardware Address. The table contains one entry with the state 'RESLV'.

```
xinu@xinu-develop-end: ~  
File Edit View Search Terminal Help  
xsh $ arp  
ARP cache:  
State Pid IP Address Hardware Address  
-----  
RESLV 10. 0. 3. 2 52:54:00:12:35:82  
xsh $
```

Menampilkan tabel pemetaan alamat jaringan (ARP Cache) yang ada di dalam sistem operasi

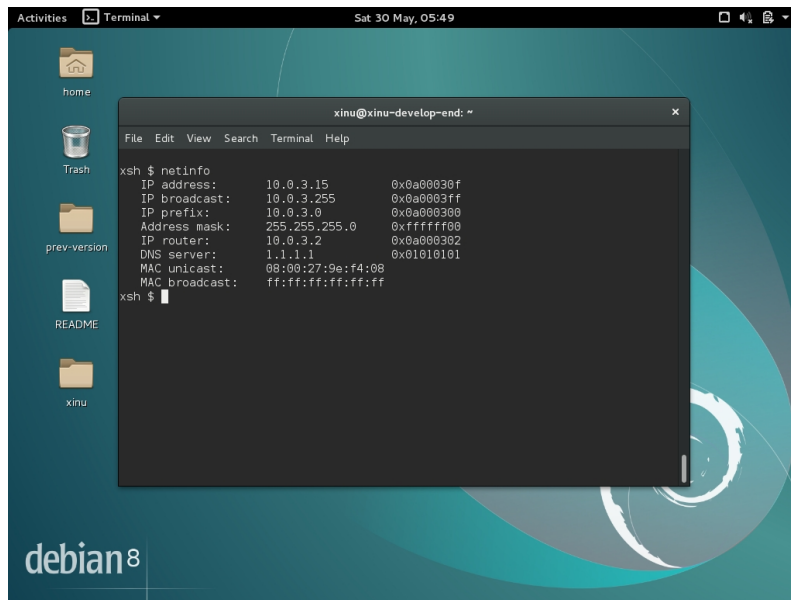
9. devdump



```
xinu@xinu-develop-end: ~  
File Edit View Search Terminal Help  
xsh $ devdump  
Device Name Minor  
-----  
0 CONSOLE 0  
1 NULLDEV 0  
2 ETHER0 0  
3 NAMESPACE 0  
4 RDISK 0  
5 RAM0 0  
6 RFILESYS 0  
7 RFILE0 0  
8 RFILE1 1  
9 RFILE2 2  
10 RFILE3 3  
11 RFILE4 4  
12 RFILE5 5  
13 RFILE6 6  
14 RFILE7 7  
15 RFILE8 8  
16 RFILE9 9  
17 LFILESYS 0  
18 LFILE0 0  
19 LFILE1 1  
20 LFILE2 2
```

Memantau dan mencetak status atau informasi detail dari perangkat keras

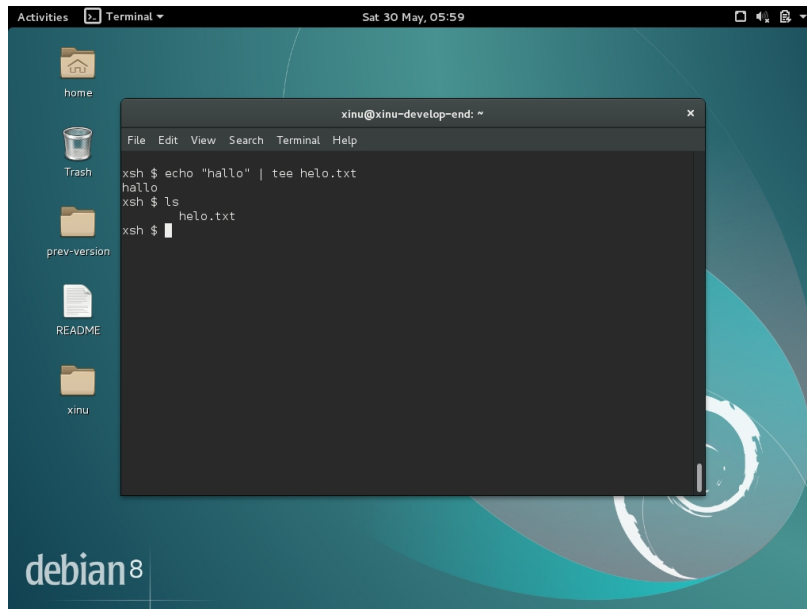
10. netinfo



```
xinu@xinu-develop-end: ~  
File Edit View Search Terminal Help  
xsh $ netinfo  
IP address: 10.0.3.15 0x0a00030f  
IP broadcast: 10.0.3.255 0x0a0003ff  
IP prefix: 10.0.3.0 0x0a000300  
Address mask: 255.255.255.0 0xffffffff00  
IP router: 10.0.3.2 0x0a000302  
DNS server: 1.1.1.1 0x01010101  
MAC unicast: 08:00:27:9e:f4:08  
MAC broadcast: ff:ff:ff:ff:ff:ff  
xsh $
```

Menampilkan informasi konfigurasi dan status jaringan pada sistem secara real-time.

11. echo dan tee



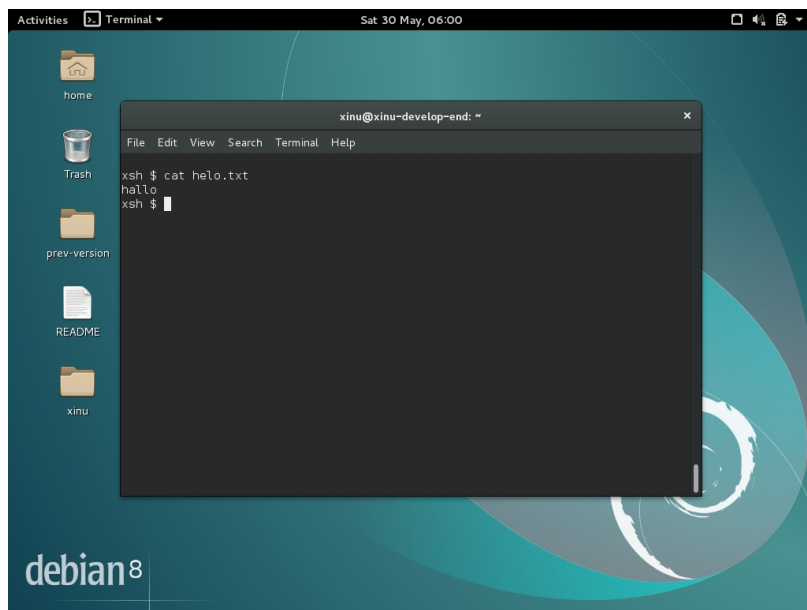
A terminal window titled 'xinu@xinu-develop-end: ~' is open on a Debian 8 desktop. The terminal shows the following commands and output:

```
xsh $ echo "hallo" | tee helo.txt
hallo
xsh $ ls
xsh $
```

The desktop background is teal with a white swirl. The left sidebar contains icons for 'home', 'Trash', 'prev-version', 'README', and 'xinu'. The terminal window has a menu bar with 'File', 'Edit', 'View', 'Search', 'Terminal', and 'Help'.

echo menampilkan teks ke dalam terminal, tee menyimpan teks yang telah kita ketik ke dalam file yang telah kita tentukan namanya.

12. Cat



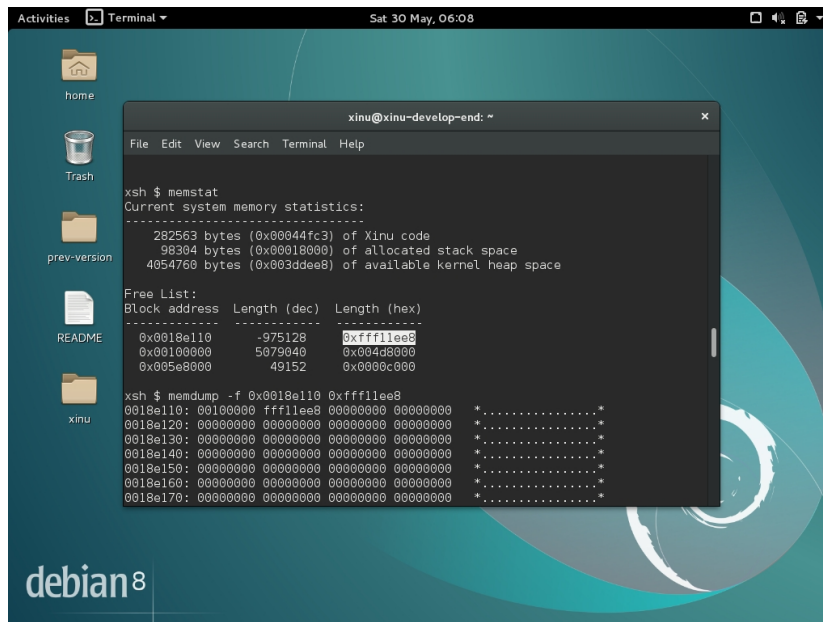
A terminal window titled 'xinu@xinu-develop-end: ~' is open on a Debian 8 desktop. The terminal shows the following commands and output:

```
xsh $ cat helo.txt
hallo
xsh $
```

The desktop background is teal with a white swirl. The left sidebar contains icons for 'home', 'Trash', 'prev-version', 'README', and 'xinu'. The terminal window has a menu bar with 'File', 'Edit', 'View', 'Search', 'Terminal', and 'Help'.

Melihat isi dari sebuah file

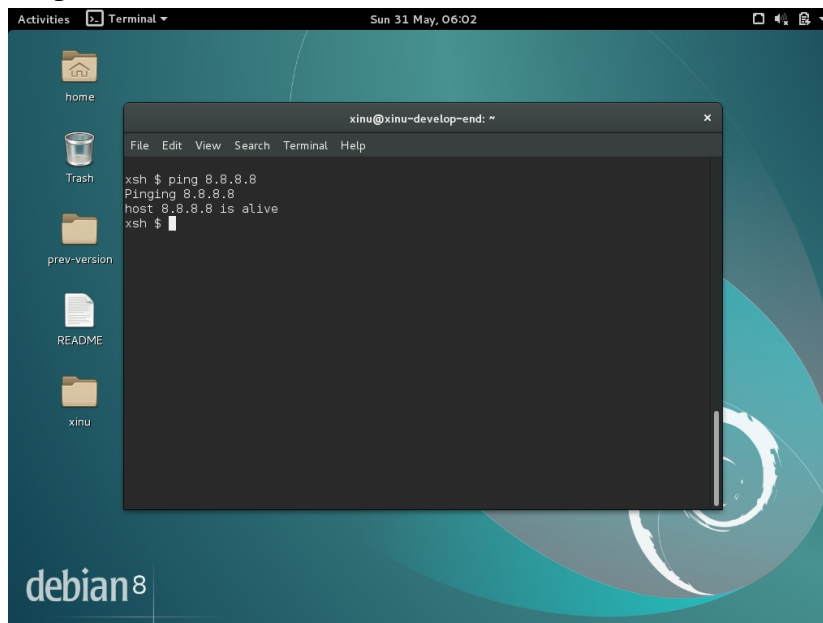
13. Memstat dan memdump



```
xinu@xinu-develop-end: ~  
File Edit View Search Terminal Help  
xsh $ memstat  
Current system memory statistics:  
-----  
282563 bytes (0x00044fc3) of Xinu code  
98304 bytes (0x00018000) of allocated stack space  
4054768 bytes (0x003ddee8) of available kernel heap space  
  
Free List:  
Block address Length (dec) Length (hex)  
-----  
0x0018e110 -975128 0xffff1ee8  
0x0018e000 5079040 0x004d0000  
0x005e0000 49152 0x0000c000  
  
xsh $ memdump -f 0x0018e110 0xffff1ee8  
0018e110: 00100000 fff1ee00 00000000 00000000 *.....*  
0018e120: 00000000 00000000 00000000 00000000 *.....*  
0018e130: 00000000 00000000 00000000 00000000 *.....*  
0018e140: 00000000 00000000 00000000 00000000 *.....*  
0018e150: 00000000 00000000 00000000 00000000 *.....*  
0018e160: 00000000 00000000 00000000 00000000 *.....*  
0018e170: 00000000 00000000 00000000 00000000 *.....*
```

Memstat memberikan ringkasan statistik memori secara garis besar dalam bentuk table, sedangkan Memdump memberikan rincian teknis atau membongkar isi memori tersebut.

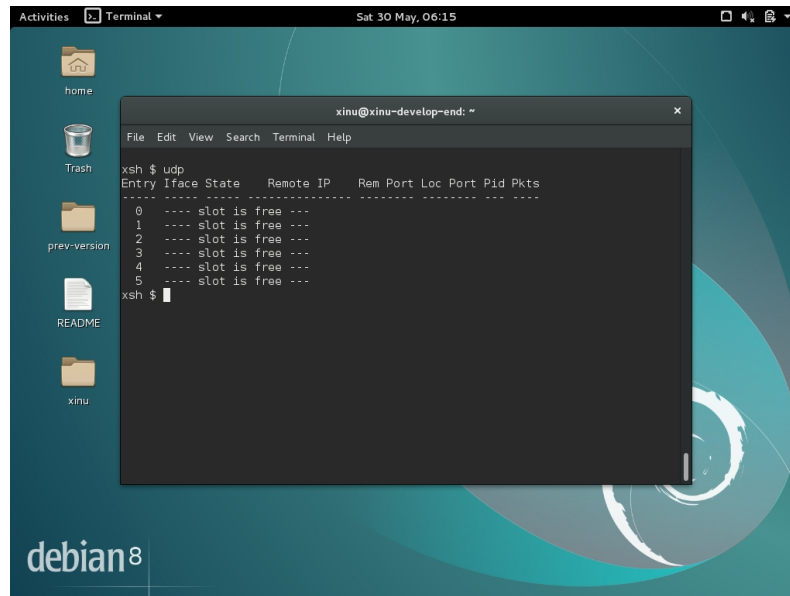
14. Ping



```
xinu@xinu-develop-end: ~  
File Edit View Search Terminal Help  
xsh $ ping 8.8.8.8  
Pinging 8.8.8.8  
host 8.8.8.8 is alive  
xsh $
```

Menguji dan memverifikasi konektivitas jaringan antara perangkat Anda dan host atau server tujuan.

15. Udp

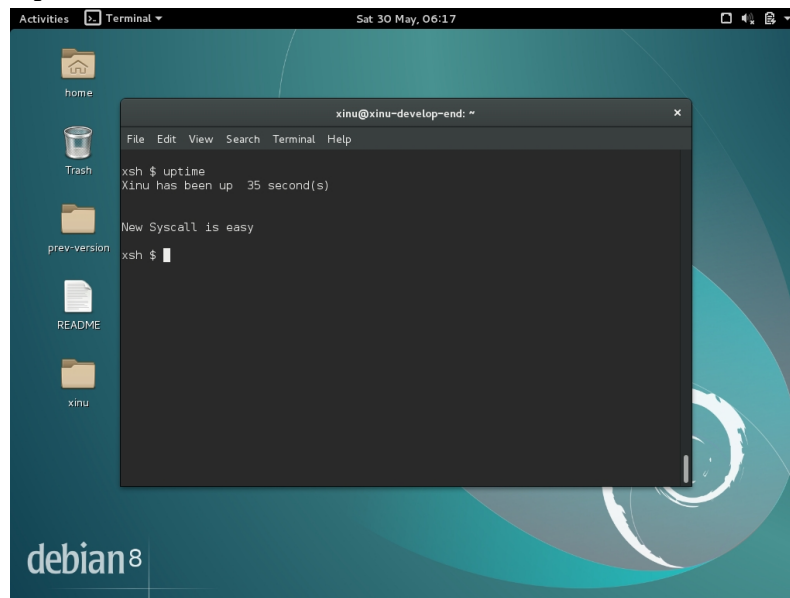


A terminal window titled 'xinu@xinu-develop-end: ~' is open on a Debian 8 desktop. The user has entered the command 'xsh \$ udp'. The output shows a table of UDP entry information:

```
xsh $ udp
Entry Iface State Remote IP Rem Port Loc Port Pid Pkts
-----
0 ---- slot is free ---
1 ---- slot is free ---
2 ---- slot is free ---
3 ---- slot is free ---
4 ---- slot is free ---
5 ---- slot is free ---
xsh $
```

Mengirimkan paket data UDP (User Datagram Protocol) dari sistem Xinu Anda ke mesin atau perangkat lain di dalam jaringan.

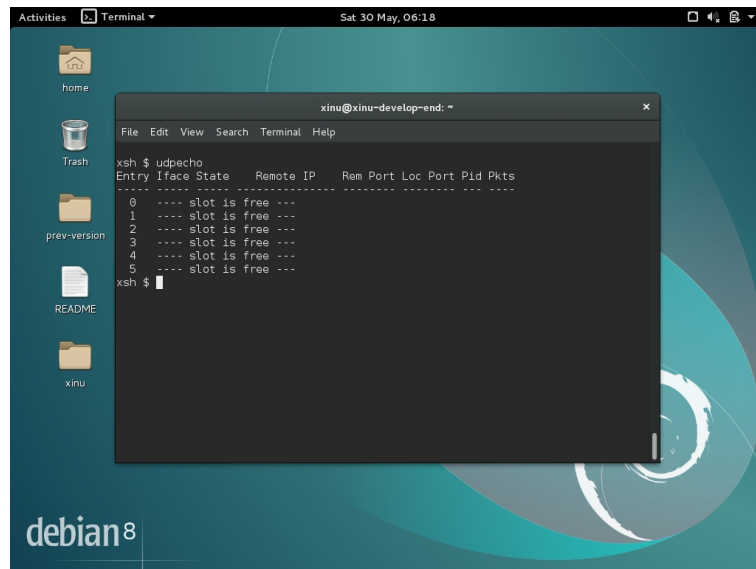
16. Uptime



A terminal window titled 'xinu@xinu-develop-end: ~' is open on a Debian 8 desktop. The user has entered the command 'xsh \$ uptime'. The output shows the system uptime:

```
xsh $ uptime
Xinu has been up 35 second(s)
New Syscall is easy
xsh $
```

17. Udpecho

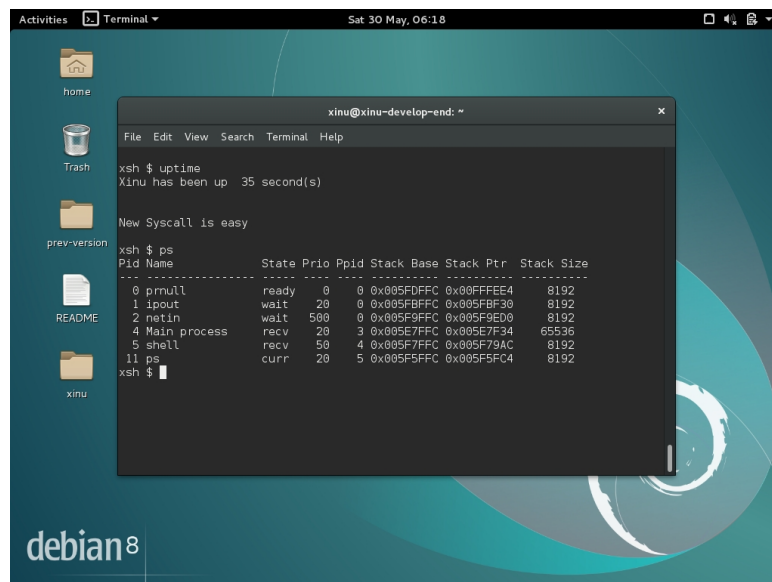


A terminal window titled 'xinu@xinu-develop-end: ~' is open on a Debian 8 desktop. The window shows the output of the 'xsh \$ udpcho' command. The output is a table with columns: Entry, Iface, State, Remote IP, Rem Port, Loc Port, Pid, and Pkts. The table shows six entries, all with the state 'slot is free'.

```
xinu@xinu-develop-end: ~  
File Edit View Search Terminal Help  
xsh $ udpcho  
Entry Iface State Remote IP Rem Port Loc Port Pid Pkts  
-----  
0 ---- slot is free ---  
1 ---- slot is free ---  
2 ---- slot is free ---  
3 ---- slot is free ---  
4 ---- slot is free ---  
5 ---- slot is free ---  
xsh $
```

Mengirimkan sebuah paket UDP ke suatu echo server

18. Ps



A terminal window titled 'xinu@xinu-develop-end: ~' is open on a Debian 8 desktop. The window shows the output of the 'xsh \$ ps' command. The output is a table with columns: Pid, Name, State, Prio, Ppid, Stack, Base, Stack Ptr, and Stack Size. The table shows several processes, including 'prnull', 'ipout', 'netin', 'Main process', 'shell', and 'ps'.

```
xinu@xinu-develop-end: ~  
File Edit View Search Terminal Help  
xsh $ uptime  
Xinu has been up 35 second(s)  
  
New Syscall is easy  
xsh $ ps  
Pid Name State Prio Ppid Stack Base Stack Ptr Stack Size  
-----  
0 prnull ready 0 0 0x005FDFFC 0x005FEEA4 8192  
1 ipout wait 20 0 0x005FDFFC 0x005F8F00 8192  
2 netin wait 500 0 0x005F9FFC 0x005F8ED0 8192  
4 Main process recv 20 3 0x005E7FFC 0x005E7F34 65536  
5 shell recv 50 4 0x005F7FFC 0x005F79AC 8192  
11 ps curr 20 5 0x005F5FFC 0x005F5FC4 8192  
xsh $
```

Menunjukkan process yang sedang berjalan

19. Sleep dan kill

```

xinu@xinu-develop-end: ~
File Edit View Search Terminal Help
4 ---- slot is free ---
5 ---- slot is free ---
xsh $ sleep 100 &
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE4 8192
1 ipout       wait   20   0 0x005FBFFC 0x005FBF30 8192
2 netin       wait  500   0 0x005F9FFC 0x005F9ED0 8192
4 Main process recv   20   3 0x005E7FFC 0x005E7F34 65536
5 shell       recv   50   4 0x005F7FFC 0x005F79AC 8192
15 sleep      sleep  20   5 0x005F5FFC 0x005F5F18 8192
16 ps        curr   20   5 0x005F3FFC 0x005F3FC4 8192
xsh $ kill 15
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE4 8192
1 ipout       wait   20   0 0x005FBFFC 0x005FBF30 8192
2 netin       wait  500   0 0x005F9FFC 0x005F9ED0 8192
4 Main process recv   20   3 0x005E7FFC 0x005E7F34 65536
5 shell       recv   50   4 0x005F7FFC 0x005F79AC 8192
18 ps        curr   20   5 0x005F5FFC 0x005F5FC4 8192
xsh $
  
```

Sleep digunakan menunda eksekusi suatu proses selama durasi tertentu, kill digunakan untuk menghentikan suatu process

Jawaban Soal No. 2 :

- Berapa jumlah perintah pada Xinu?
Ada 23 perintah pada xinu argecho, date, ls, ns, sleep, udpserver , arp, devdump, kill, netinfo, tee, cat, echo, memdump, ping, udp, uptime, clear, help, memstat, ps, udpecho, ?
- Sebutkan 2 perintah yang mempunyai fungsi yang sama!
help dan ? memiliki fungsi yang sama, yaitu untuk menampilkan panduan daftar perintah.
- Berapa IP address Xinu?

```

xsh $ netinfo
IP address:      10.0.3.15          0x0a00030f
IP broadcast:    10.0.3.255       0x0a0003ff
IP prefix:       10.0.3.0         0x0a000300
Address mask:    255.255.255.0    0xffffffff00
IP router:       10.0.3.2         0x0a000302
DNS server:      1.1.1.1          0x01010101
MAC unicast:     08:00:27:9e:f4:08
MAC broadcast:   ff:ff:ff:ff:ff:ff
  
```

10.0.3.15 (hasil output netinfo)

- Perintah apa yang digunakan untuk mengetahui IP address?
Perintah yang digunakan adalah netinfo.
- Berapa IP DNS server yang digunakan oleh Xinu?

1.1.1.1 (hasil output netinfo).

- f. Terdapat berapa proses yang sedang berjalan pada Xinu?

```
xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr Stack Size
-----
 0 prnull         ready  0    0 0x005FDFFC 0x00FFFE4 8192
 1 ipout          wait  500  0 0x005FBFFC 0x005FBF30 8192
 2 netin          wait  500  0 0x005F9FFC 0x005F9ED0 8192
 4 Main process   recv  20   3 0x005E7FFC 0x005E7F34 65536
12 shell         recv  20   4 0x005F7FFC 0x005F79AC 4096
14 ps            curr  20  12 0x005F6FFC 0x005F6DB0 8192
xsh $
```

Terdapat 6 process yang sedang berjalan.

- g. Proses apa yang mempunyai prioritas paling rendah?
Prnull dengan nilai prioritas 0.
- h. Proses apa yang mempunyai ukuran paling besar?
Main process dengan stacksize 65536.
- i. Proses apa yang berada dalam state current?
Proses ps.
- j. Proses apa yang berada dalam state suspend?
Process ipout dan netin sedang dalam process waiting.
- k. Berapa PID (Process ID) dari Main process?
PID dari main process adalah 4.
- l. Berapa PPID (Parent Process ID) dari Main process?
PPID dari main process adalah 3.
- m. Terminasi proses ps!

```
xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr Stack Size
-----
 0 prnull         ready  0    0 0x005FDFFC 0x00FFFE4 8192
 1 ipout          wait  500  0 0x005FBFFC 0x005FBF30 8192
 2 netin          wait  500  0 0x005F9FFC 0x005F9ED0 8192
 4 Main process   recv  20   3 0x005E7FFC 0x005E7F34 65536
12 shell         recv  20   4 0x005F7FFC 0x005F79AC 4096
14 ps            curr  20  12 0x005F6FFC 0x005F6DB0 8192
xsh $ kill 14
```

- n. kill: cannot kill process 14
Process ps tidak dapat di terminasi karena merupakan sebuah proses sementara dimana setelah selesai eksekusi, PID nya sudah tidak valid saat perintah kill dieksekusi.
- o. Terminasi proses shell!

```

 4 Main process   recv    20    3 0x005E7FFC 0x005E7F34 65536
12 shell         recv    20    4 0x005F7FFC 0x005F79AC 4096
17 ps           curr    20   12 0x005F6FFC 0x005F6DA4 8192
xsh $ kill 12

```

```

Main process recreating shell

```

```

-----
XINU
-----

```

```

Welcome to Xinu!

```

p. xsh \$ █

q. Berapa lama Xinu sudah berjalan?

```

xsh $uptime
Xinu has been up  3 minute(s)  52 second(s)

```

r.

s. Perintah apa yang digunakan untuk membersihkan tampilan terminal?

t. Perintah yang digunakan untuk membersihkan tampilan terminal adalah "clear".

u. Perintah apa yang digunakan untuk menampilkan statistik memory pada Xinu?
Perintahnya adalah memstat.

v. Lakukan ping ke DNS server!

```

xsh $ ping 1.1.1.1
Pinging 1.1.1.1
host 1.1.1.1 is alive
xsh $ █

```

w. Lakukan perintah untuk menampilkan "HELLO WORLD!" pada terminal!

```

xsh $ echo HELLO WORLD!
HELLO WORLD!
xsh $ █

```

x. Silahkan keluar dari terminal Xinu!

